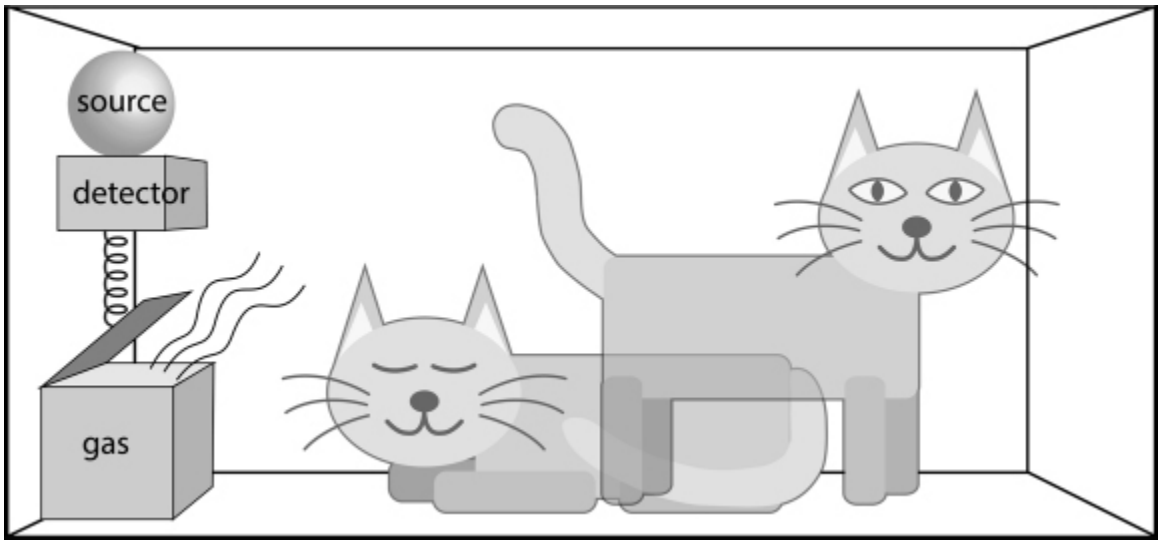


As far as that goes, he was right. Once we have an Everettian perspective on quantum dynamics, we accept that the wave function smoothly evolves into an equal superposition of two possibilities, one in which the cat is asleep and the other in which it is awake. But decoherence tells us that the cat is also entangled with its environment, consisting of all the air molecules and photons within the box. The effective branching into separate worlds happens almost right away after the detector clicks. By the time the experimenter gets around to opening the box, there are two branches of the wave function, each of which has a single cat and a single experimenter, not a superposition.

This solves Schrödinger's original worry, but raises another one. Why is it that when we open the box, the particular decohered quantum states we see are either that of an awake cat, or an asleep cat? Why don't we see some superposition of both? "Awake" and "Asleep" together represent just one possible basis for the cat system, just as "spin-up" and "spin-down" do for the electron. Why is that basis preferred over any other one?

The physical process that matters is stuff in the environment—gas molecules, photons—interacting with the physical system under consideration. Whether a particular particle actually does interact with the cat will depend on where the cat is. A given photon might very well be absorbed by a cat that is awake and prowling around the box, but completely miss a cat that is sleeping on the floor.